Is every resident within 3 minutes of a public EV charger?

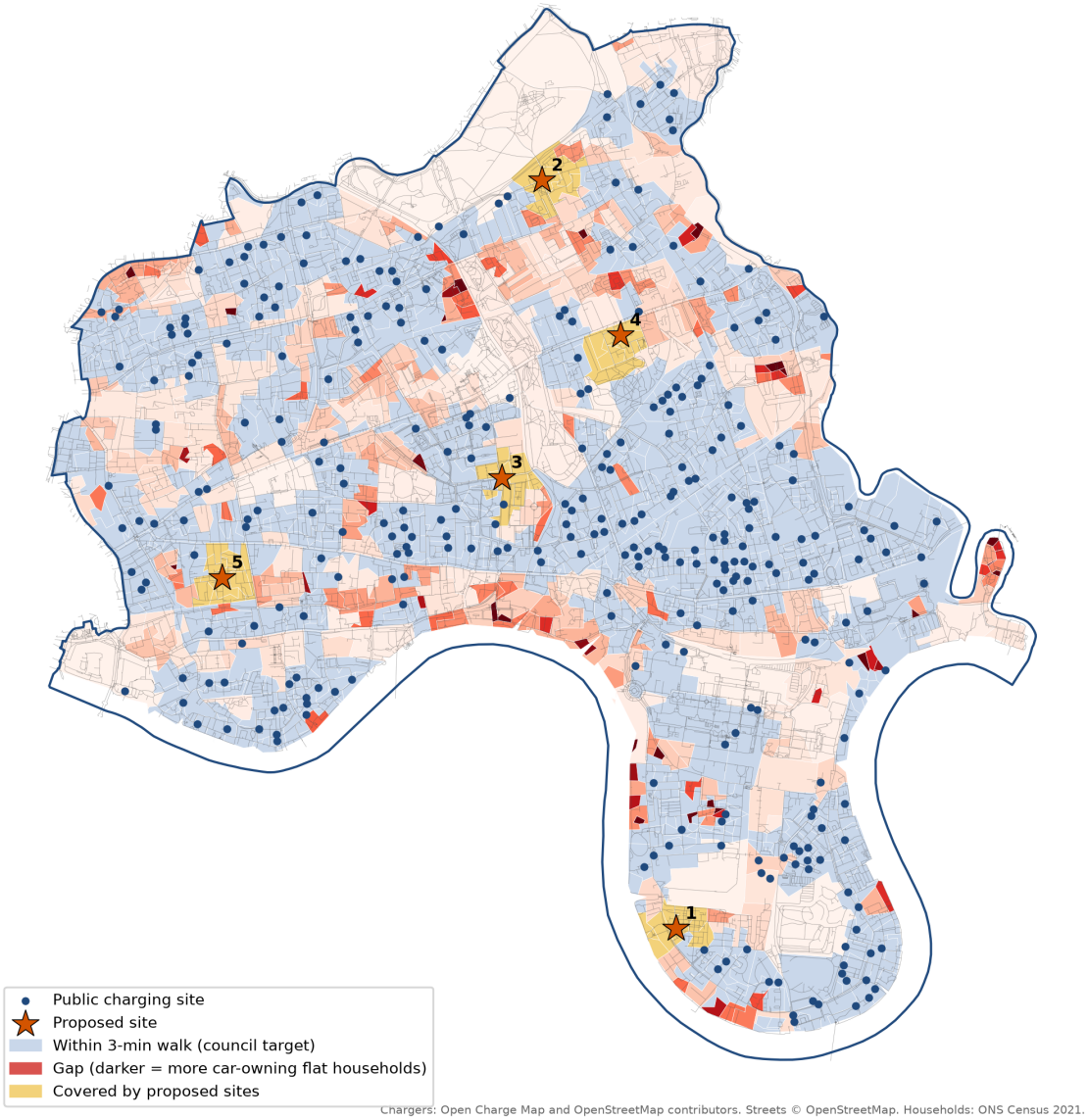
Testing a Tower Hamlets charging ambition with open data

Why this question? In Tower Hamlets, 93% of homes have no off-street parking, so most drivers who switch to an electric car will rely on public chargers. The council has installed around 2,000 lamp-column charge points and set an ambition that no resident should be more than a 3-minute walk from one. This analysis tests that ambition for the households that depend on it most: those who own a car but live in a flat.

62%
of car-owning flat households are within a 3-minute walk of a public charger

88%
are within a 5-minute walk

≈13,000
households still beyond the 3-minute target, typically by 1–4 minutes

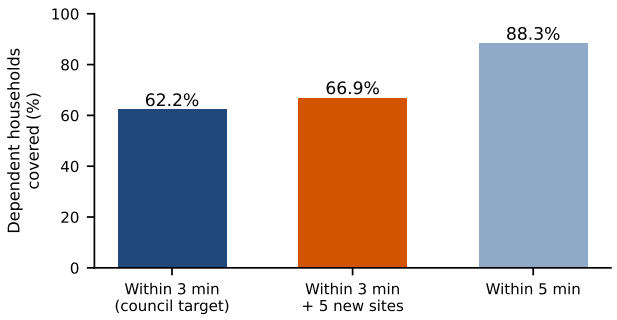


Census 2021 Output Areas (913 areas, 120,554 households). Red areas are beyond a 3-minute walk (about 240 m) along the street network; darker red means more car-owning households living in flats. Chargers within about 500 m of the boundary count towards coverage.

Who is left out

Of an estimated 34,300 households that depend on public charging, about 13,000 are more than a 3-minute walk from a charger. The gaps are scattered small pockets, mostly along the riverside and on parts of the Isle of Dogs, rather than a single unserved area.

The 15 largest gaps were checked on a live charger map. In 13 of them the nearest charger really was beyond 3 minutes, usually around a 5-minute walk and at most 12 minutes. In the other two, a closer charger existed that open data had missed.



So what

- **The 3-minute ambition is demanding; at 5 minutes, coverage is far higher.** Coverage rises from 62% to 88% when the threshold moves from 3 to 5 minutes, so most remaining gaps are near-misses rather than charging deserts.
- **Closing the last gaps will take many small additions.** Five optimally placed new sites would reach about 1,600 more households, lifting 3-minute coverage only from 62% to 67%. Each further site helps fewer people, because the gaps are scattered.
- **Target the pockets, not the borough.** The gap areas listed in the project's `gap_check.csv` are a ready-made shortlist for checking on the ground before further installations.
- **Publishing locations would make the target measurable.** Open datasets list 452 charge points in the borough against about 2,000 reported by the council. Their locations appear largely complete, but an official open dataset of charger locations would let residents and analysts track progress directly.

Method

- **Who depends on public charging:** households with a car (Census 2021, TS045) multiplied by each area's share of households in flats (TS044), for 913 Output Areas.
- **Chargers:** 315 charging sites in the borough from Open Charge Map and OpenStreetMap, merged and de-duplicated within 30 m, plus sites just outside the boundary.
- **Walking distance:** shortest paths on the OpenStreetMap walking network; 3 minutes at 4.8 km/h is about 240 m, with 5 minutes (400 m) as a sensitivity check.
- **New sites:** candidate on-street locations every 100 m along residential streets, chosen by a greedy search that adds, at each step, the site reaching the most uncovered dependent households.
- **Tools:** Python (GeoPandas, OSMnx, NetworkX, Matplotlib); code in this repository.

Data quality and limitations

- **Open data checked against a live map.** OpenStreetMap alone showed only 19 sites in the borough; adding Open Charge Map raised this to 315. Checking the 15 largest gaps confirmed 13, so the 62% figure is reliable but probably a slight underestimate.
- **An estimated population.** The Census does not cross car ownership with dwelling type for small areas, so car ownership is assumed to be spread evenly across housing types within each Output Area. The data also dates from 2021.
- **Access, not adequacy.** The analysis measures distance only. It does not consider charger speed, price, availability at busy times, or whether a bay is free.
- **Walking times are approximate.** Each area's households are placed at a single point, and walking speed is assumed constant.

Independent analysis using public data. Chargers: Open Charge Map and OpenStreetMap contributors (ODbL). Households: ONS Census 2021, Open Government Licence. Council figures from Tower Hamlets Council announcements (2025–2026).